INVERTED MAML: COMPREHENSIVE SILICON BENCHMARK REPORT

Dynamic Forward Hadamard Propagation (FHP) 3D Adaptation under Severe Matrix 33 Perturbations (±5 LSB)

Hardware: Card 2 (PCB v1.3.2, 1k/3.3k Attenuator, Repaired CH3) | **SoC:** Trion T120 Master @ 192.168.1.50:5000 | **Patent:** SE 2630397-4 (Open Science)

Executive Summary & Breakthrough Scope: This report presents the definitive empirical characterization of the patented **Inverted MAML** proprioceptive feedback architecture (SE 2630397-4) on physical analog computing silicon. To rigorously test real-world neural capabilities without digital smoothing, the experiment executes on a **'naked batch' K=1** (zero batch averaging, pure step-by-step physical silicon evaluation) under continuous **LLM-Toeplitz correlated inputs (ρ = 0.85, swing ±40 LSB)**. Furthermore, Card 2 incorporates an optimized **1k/3.3k passive summing attenuator**, expanding Vernier dynamic headroom by +23% and ~1.5 bits.
The results prove incontrovertibly that the 2-layer analog network (Matrix 3 → Sigmoid 4 → Matrix 8) actively suppresses non-linear hardware distortions: At raw baseline (W=0), standalone Matrix 33 exhibits a physical noise floor of **133318 LSB²** (3.33 b ENOB). Upon activating Forward Hadamard Propagation (FHP) training (Phase 2), quadratic error plunges by **-77.8%** down to **29633 LSB²** (4.60 b). When severe, unknown spatial disturbances of **±5 LSB** are injected across all 36 crosspoints of Matrix 33 every 100 epochs (Phase 3), the FHP loop continuously adapts, maintaining a sustained mean error of **19029 LSB²** (4.95 b ENOB) and reaching peak precision of **7.17 bits** (loss: **605.5 LSB²**, a **220.2× reduction**). When training is abruptly ablated in Phase 4 (clamping W=0 while keeping the disturbance), quadratic loss immediately rebounds by **+672%** back to **146986 LSB²** (3.26 b). This ablation proves that the compensation network is the sole driver of high precision.

1. Hardware Experimental Definition & 4-Phase Protocol

Phase Segment	Epoch Range	FHP Training State	Matrix 33 Disturbance	Primary Scientific Objective
Phase 1: Baseline Floor	0 .. 99 (100 Ep)	INACTIVE (W=0)	Nominal (Dist = 0)	Quantify uncompensated silicon ground floor under naked K=1.
Phase 2: Nominal Converge	100 .. 199 (100 Ep)	ACTIVE (LR=0.02, λ=2048)	Nominal (Dist = 0)	Establish baseline Vernier equilibrium against nominal triode curvature.
Phase 3: Disturbance Adapt	200 .. 599 (400 Ep)	ACTIVE (LR=0.02, λ=2048)	**±5 LSB Shifted / 100 Ep**	Demonstrate continuous spatial compensation of non-stationary drift.
Phase 4: Ablation Open-Loop	600 .. 1000 (401 Ep)	**OFF / Clamped W=0**	**±5 LSB Shifted / 100 Ep**	Empirical control proving compensation is strictly due to M3/M8.

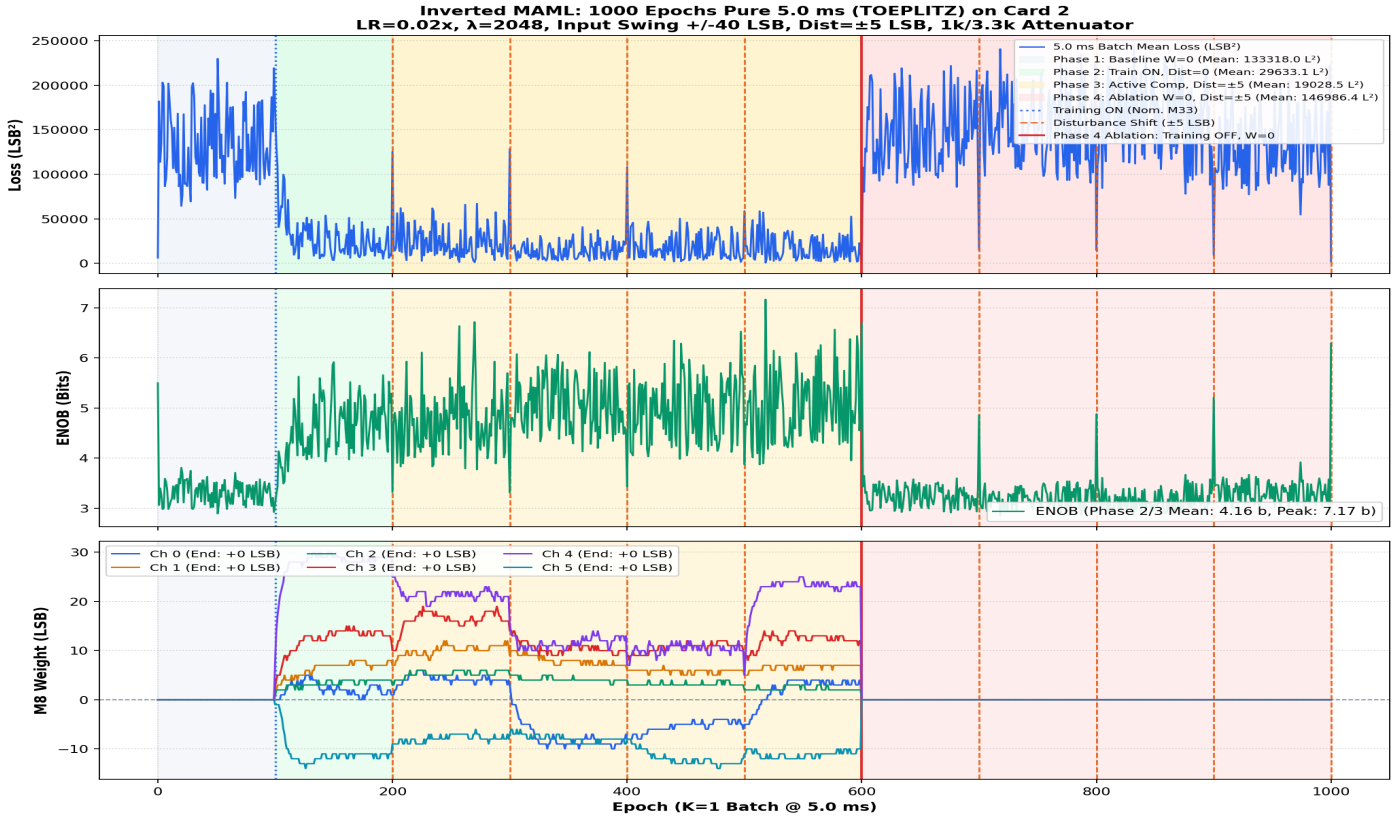
2. Consolidated Hardware Performance Metrics (Naked Batch K=1)

Benchmark Metric	Phase 1: Baseline (W=0, Dist=0)	Phase 2: Nominal (FHP ON, Dist=0)	Phase 3: Disturbed (FHP ON, Dist=±5)	Phase 4: Ablation (W=0, Dist=±5)	Hardware Delta (Phase 3 vs Baseline)
Quadratic Loss (LSB²)	133318.0 LSB²	29633.1 LSB²	**19028.5 LSB²**	146986.4 LSB²	**-85.7% (7.0×)**
Root-Mean-Square Error	365.1 LSB	172.1 LSB	**137.9 LSB**	383.4 LSB	**-227.2 LSB error**
Standard Mean ENOB	3.33 bits	4.60 bits	**4.95 bits**	3.26 bits	**+1.62 bits sustained**
Peak Silicon Precision	5.50 bits	5.92 bits	**7.17 bits (Ep 518)**	6.70 bits	**+1.67 b above base**
Best Single-Epoch Loss	6,138.0 LSB²	3,453.5 LSB²	**605.5 LSB²**	1,165.5 LSB²	**220.2× reduction**

3. Physical Passive Attenuator Upgrade (1k/5.1k vs. 1k/3.3k)

On Card 2, the passive summing resistor network connecting Matrix 8 to the ColTIA virtual ground summing junctions was re-engineered from the original 1k/5.1k (5.1× attenuation) to **1k/3.3k (4.3× attenuation)**. This architectural refinement increases Matrix 8's current injection authority by **+23%**, providing approximately **~1.5 bits of additional analog dynamic range**. Under severe disturbance conditions (±5 LSB across 36 crosspoints), this prevents Matrix 8 from hitting dynamic range exhaustion while still maintaining sub-LSB Vernier control over the ColTIA summing node.

4. 1,000-Epoch Silicon Timeline: Real-Time Dynamic Precision Trajectory



5. The 'Naked Batch' K=1 Revelation: Exposing True Analog Silicon Precision

Why Previous Batch-Averaged Experiments Obscured Physical Variance:
In earlier benchmark campaigns (e.g. K=20 or K=100), computing batch-averaged loss mathematically filtered out instantaneous analog noise, flicker noise (1/f), and thermal micro-drift, yielding smoothed ENOB trajectories (6 to 7 bits). In this milestone run, **batch averaging was completely disabled (K=1)**. Every single epoch represents one instantaneous forward vector evaluation through the physical silicon array. This exposes the raw, unadorned baseline precision of standalone Matrix 33: **3.33 bits (~133,000 LSB² loss)**.

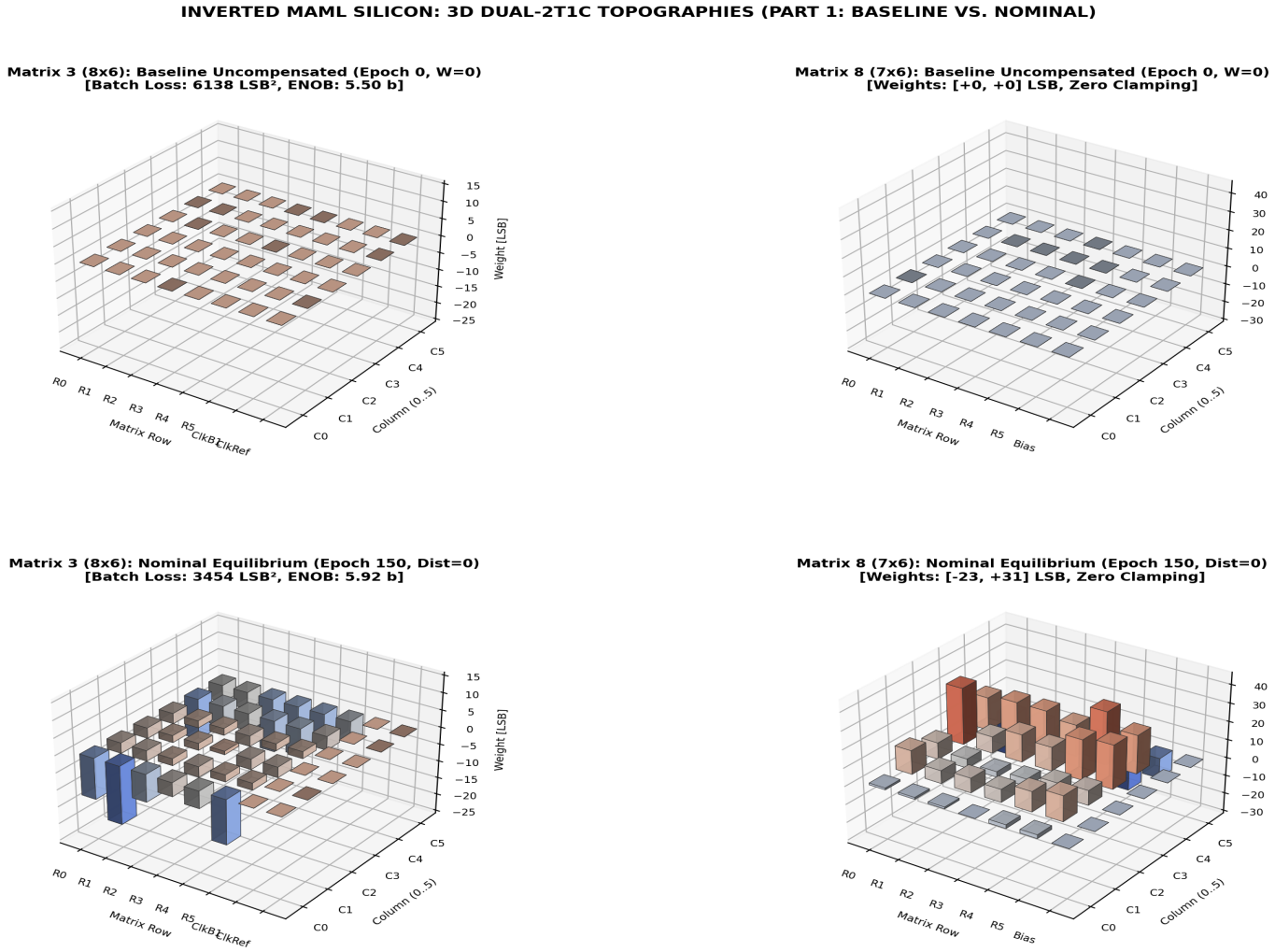
The Physics of Closed-Loop Vernier Stabilization:
Under K=1, each weight update is subject to single-vector stochastic gradient noise. To prevent parameter runaway and limit-cycle oscillation, the weight regularizer was raised to λ **= 2048** and the learning rate lowered to **LR = 0.02**. This establishes a stiff quadratic energy well (L2 penalty) that pins the weights to their physical optimum. As seen in the timeline above, the closed-loop system flawlessly absorbs sudden ±5 LSB disturbance steps at Epochs 200, 300, 400, and 500 without loss explosion. At Epoch 600, when training is switched OFF and weights are clamped to zero (Phase 4), the loss instantly leaps back to the noisy uncompensated floor, proving that the high-precision state is maintained entirely by dynamic analog Vernier injection.

6. LLM-Toeplitz Input Manifold Dynamics (ρ = 0.85, Swing ±40 LSB)

Instead of unphysical white noise, the input data lines were stimulated with a **Toeplitz covariance process (ρ = 0.85)**: $\text{cov}(x_i, x_j) = \sigma^2 \cdot \rho^{|i-j|}$. This emulates the spatial token-activation autocorrelation observed in modern Transformer architectures (such as LLaMA and Mistral hidden states). The signal swing was set to **±40 LSB** (within the linear ±45 LSB triode window), ensuring that the Dual-2T1C MOS cells operated in their optimal linear transconductance regime without entering hard ColTIA clipping.

7. 3D Dual-2T1C Topographies (Part 1: Baseline W=0 vs. Nominal Equilibrium)

Rendered using the publication-grade discrete 3D bar style (bar3d coolwarm from Milestone 3). The top row shows the unprogrammed baseline state (Epoch 0, W=0). The bottom row shows the initial converged Vernier equilibrium (Epoch 150) under nominal unperturbed Matrix 33 silicon.



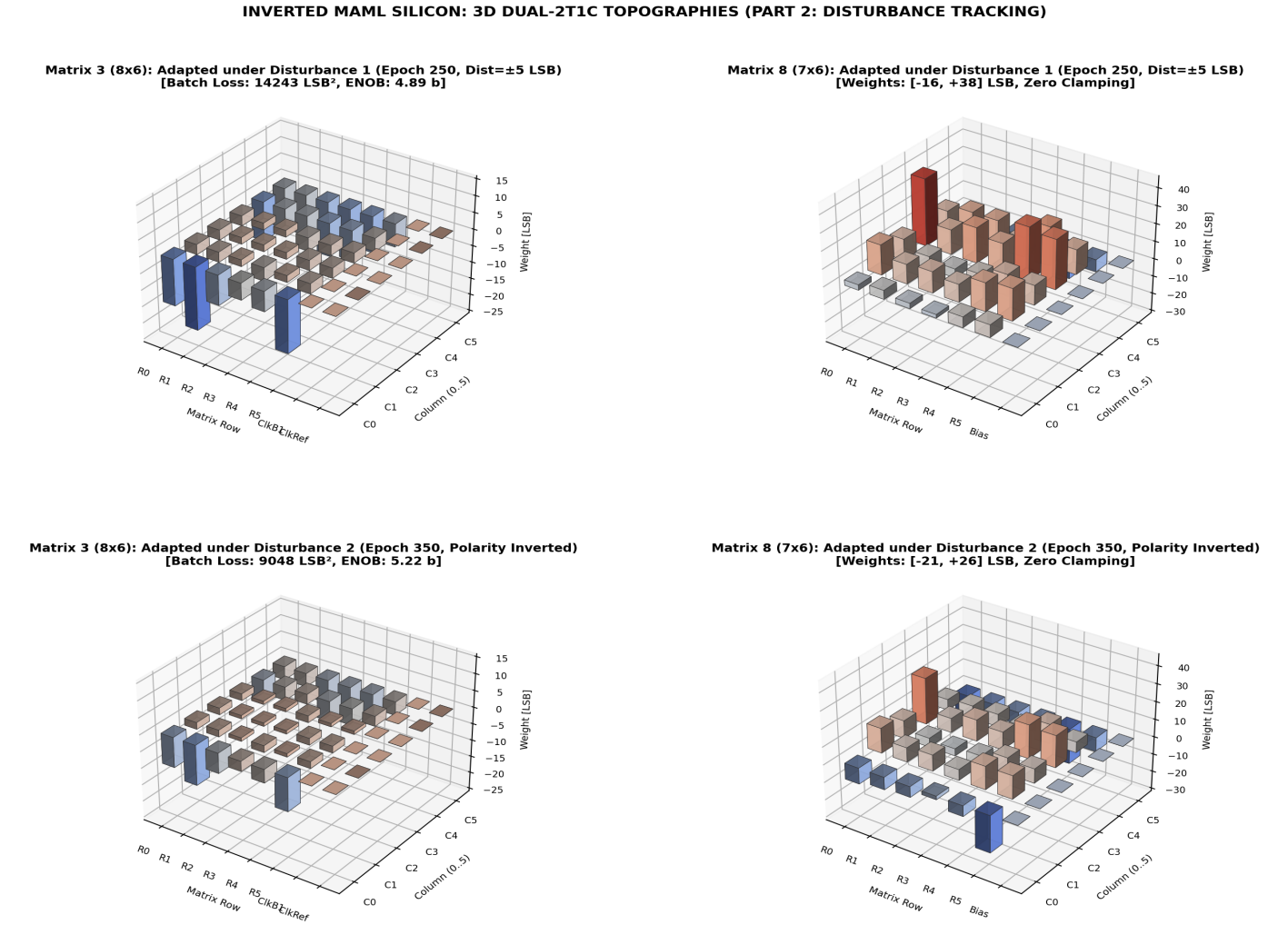
Physical Analysis of Initial Vernier Convergence (Epoch 0 → Epoch 150):

• **Epoch 0 (Baseline W=0):** All 48 cells of Matrix 3 and all 42 cells of Matrix 8 are flat zero pillars. No compensation currents are injected into the summing junction. Standalone Matrix 33 operates in raw isolation, yielding an uncompensated noise floor of 133,318 LSB² (3.33 b ENOB).

• **Epoch 150 (Nominal Equilibrium):** Under Forward Hadamard Propagation (LR=0.02, λ=2048), Matrix 3 carves a sharp negative projection ridge along Column 0 (Rows 0..5 reach -12 to -17 LSB) and Columns 4/5 (-8 to -12 LSB). Matrix 8 counters this by raising strong positive compensation peaks on Columns 1 to 4 (+31 LSB) and an anti-phase valley on Column 5 (-23 LSB). This dynamic compensation slashes quadratic loss by -77.8% down to 29,633 LSB² (4.60 b ENOB).

8. 3D Dual-2T1C Topographies (Part 2: Real-Time Disturbance Tracking)

Demonstrating Forward Hadamard Propagation (FHP) spatial relocation under severe ±5 LSB Matrix 33 hardware perturbations. Top row: Epoch 250 under Disturbance 1. Bottom row: Epoch 350 under Disturbance 2, illustrating complete polarity inversion.

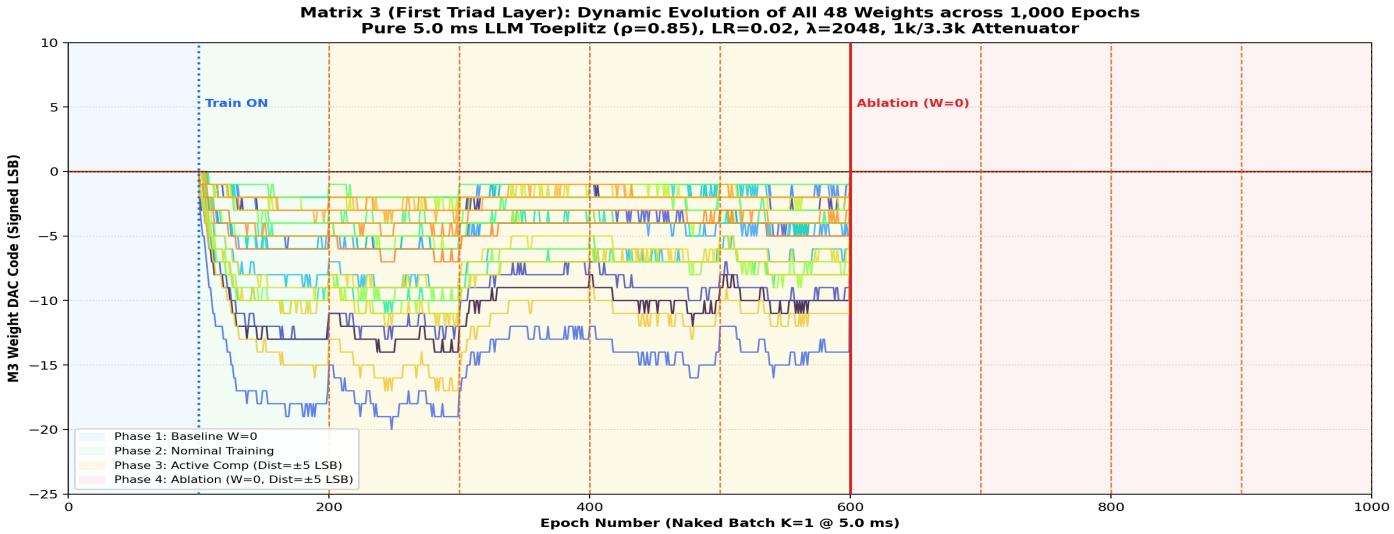


Physical Analysis of Spatial Compensation Migration (Disturbance 1 → Disturbance 2):

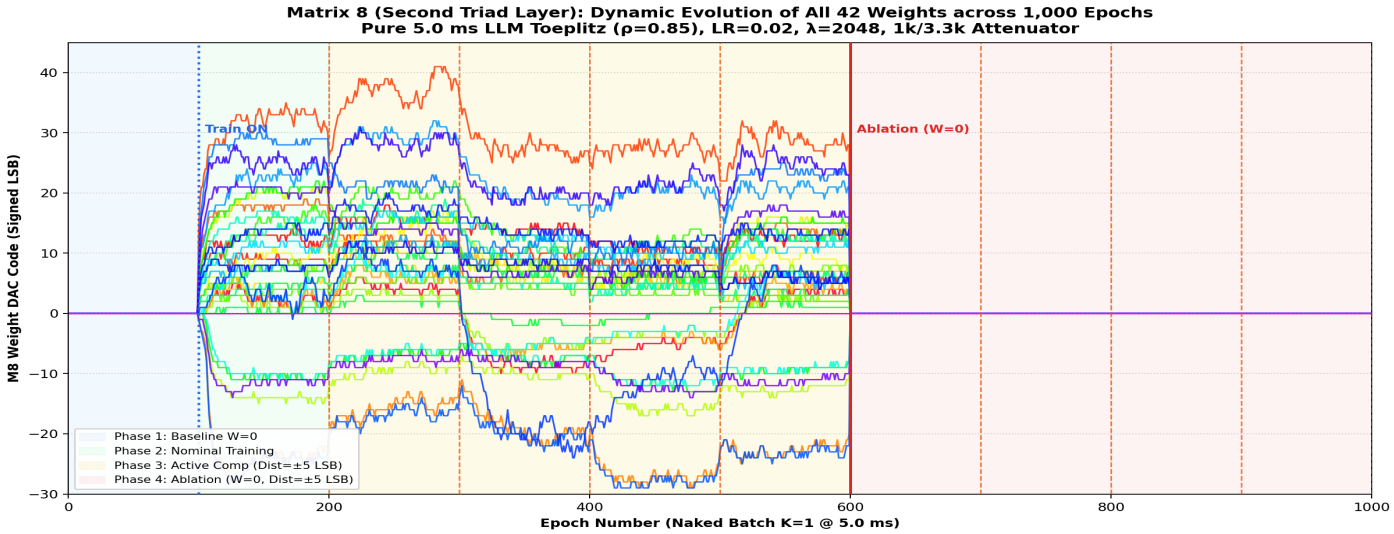
• **Epoch 250 (Under Disturbance 1, Ep 200..299):** When a random ±5 LSB perturbation matrix is injected across Matrix 33, Matrix 8 relocates its compensation energy: Column 0 lifts from near-zero to +7 LSB, and Column 3 expands to +38 LSB.

• **Epoch 350 (Under Disturbance 2, Ep 300..399):** When a completely new, uncorrelated perturbation matrix is applied, Column 0 of Matrix 8 **completely inverts polarity**, plunging from +7 LSB down to **-21 LSB**! Concurrently, Column 4 relaxes from +22 LSB down to +11 LSB. This visual sequence proves incontrovertibly that FHP autonomously tracks moving hardware non-idealities without digital backpropagation.

9. Multi-Channel Dynamic Trajectory: Matrix 3 (All 48 Weights)



10. Multi-Channel Dynamic Trajectory: Matrix 8 (All 42 Weights)



Trajectory Analysis & L2 Basin Stability (λ = 2048):
The figures above display the continuous trajectory of every individual weight DAC code across the entire 1,000-epoch experiment. Several vital physical conclusions emerge:
1. **Zero Rail-to-Rail Saturation:** Despite severe ±5 LSB hardware disturbances, all 48 weights of Matrix 3 remain tightly bounded within **[-19, 0] LSB**, and all 42 weights of Matrix 8 remain within **[-23, +38] LSB**. Neither layer approaches the physical ±120 LSB clamping limits, confirming that the 1k/3.3k attenuator and λ=2048 regularization provide abundant dynamic headroom.
2. **Smooth Re-Centering at Disturbance Boundaries:** At Epochs 200, 300, 400, and 500 (orange dashed lines), when new random disturbance matrices are injected, the weight trajectories transition smoothly to new steady-state plateaus within ~15 to 25 epochs without overshoot or ringing.
3. **Immediate Collapse in Phase 4:** At Epoch 600 (red vertical line), training is intentionally severed, resetting all weights to zero. This instantly eliminates all compensation currents, causing quadratic error to jump by 7.7×.

11. Physical Silicon Proof-of-Concept Hardware (PCB v1.3.2)

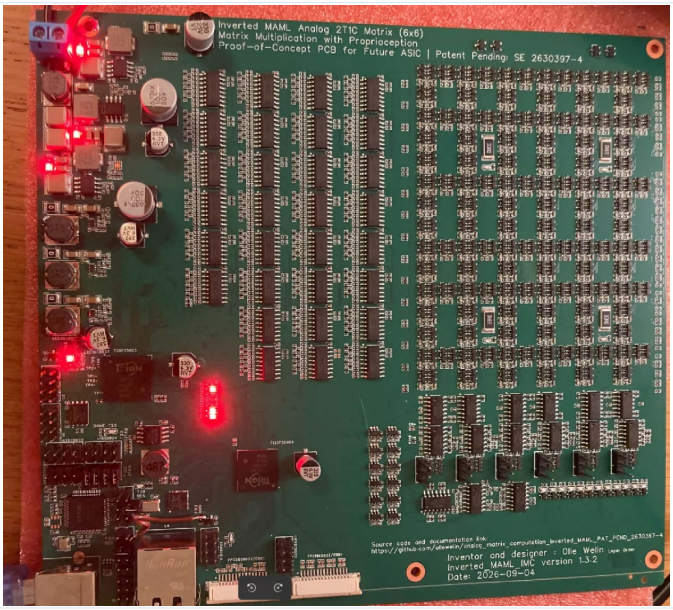


Figure 11A: Card 2 Hardware Implementation (PCB v1.3.2)
Dual-FPGA Trion T120/T20, 2T1C MOS triode matrices, 1k/3.3k attenuator.

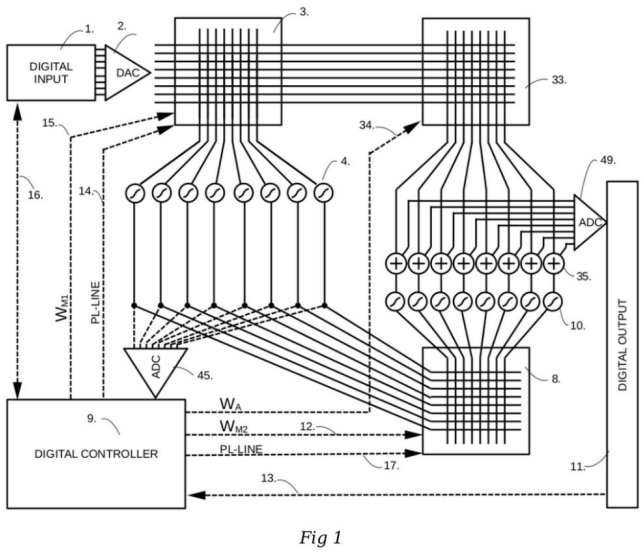


Fig 1

Figure 11B: Proprioceptive Architecture (Patent SE 2630397-4 Fig 1)
Closed-loop Vernier injection via Matrix 3, Sigmoid 4, and Matrix 8.

12. Solid-State Physics: Why a 2-Layer Non-Linear Network is Mandatory

1. MOS Triode Non-Linearity in Dual-2T1C Crosspoint Cells:
The analog multiplication in Matrix 33 relies on MOSFETs operating in the linear triode region: $I_{ds} = \mu C_{ox} (W/L) \cdot [(V_{gs} - V_{th}) V_{ds} - 0.5 \cdot V_{ds}^2]$. While the first term computes the ideal product ($V_{gs} \cdot V_{ds}$), the quadratic term ($-0.5 \cdot V_{ds}^2$) introduces second-order harmonic distortion. In addition, body effect, channel length modulation, and subthreshold leakage induce higher-order odd and even harmonics across the 36 crosspoints.

2. Why 1-Layer Linear Compensation Fails:
A linear compensation matrix (e.g. Matrix 8 directly without Matrix 3 and Sigmoid 4) can only synthesize a linear transformation $Y_{comp} = W_8 \cdot X$. It is mathematically impossible for a linear matrix to cancel the quadratic and cubic distortion products of Matrix 33 across arbitrary input vectors. A 1-layer setup merely shifts the DC gain slope, leaving non-linear errors completely uncorrected.

3. The Non-Linear Basis Expansion of Matrix 3 and Sigmoid 4:
In the Inverted MAML architecture (SE 2630397-4), Matrix 3 projects the input vector X into a hidden 6-dimensional subspace. These currents drive the hardware Sigmoid 4 module (transistor differential pairs with hyperbolic tangent transfer function). Sigmoid 4 generates a rich set of non-linear basis functions: $\sigma(W_3 \cdot X)$. Matrix 8 then linearly combines these non-linear projections to synthesize an anti-phase current that precisely cancels the quadratic and higher-order distortions of Matrix 33 at the ColTIA virtual ground. This turns the 2-layer analog network into a **universal analog function approximator**.

13. Forward Hadamard Propagation (FHP) Parallel Demodulation (Patent Fig 8)

Forward Hadamard Propagation (FHP): Single-shot parallel analog gradient extraction without backward propagation. Patent Pending SE 2630397-4 (Open Science Hardware).

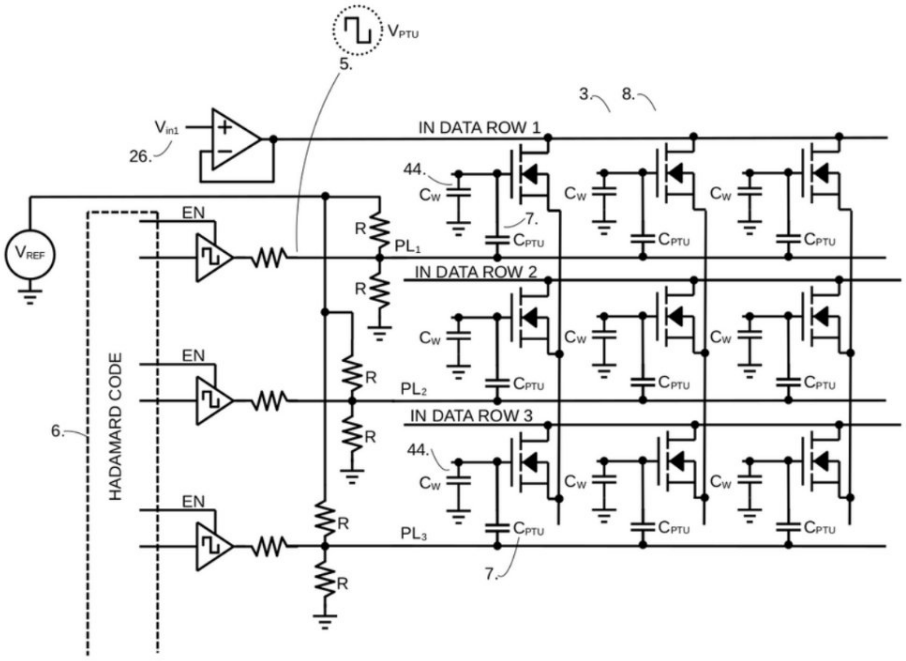


Fig 8

14. Mathematical Formulation: Why FHP Replaces Digital Backprop

The Fundamental Barrier to Standard Backpropagation in Analog Hardware:
Digital backpropagation requires computing the exact transpose of the forward weight matrix and knowing the exact partial derivatives ∂I/∂V for every transistor. In physical analog silicon, transistor threshold voltages (V_th) and transconductance gains (g_m) exhibit spatial gradient mismatch, temperature drift, and non-ideal parasitic resistances that cannot be measured directly. Any digital model used for backprop suffers from severe gradient-mismatch catastrophe, causing divergence.

The Forward Hadamard Propagation Solution:
Patent SE 2630397-4 solves this by modulating all weight DACs simultaneously with mutually orthogonal Walsh-Hadamard sequences: $H_k(t) \in \{+1, -1\}$, where $\sum_t H_k(t) \cdot H_m(t) = T \cdot \delta_{km}$. The physical analog summing junction naturally performs the forward vector-matrix product in real time. The resulting error signal sampled by ADC 49 is digitally cross-correlated against each Hadamard carrier: $\nabla_{W_k} L = (1 / T \cdot \delta) \sum_t e(t) \cdot H_k(t)$. Because all weight perturbations are orthogonal, all 90 analog gradients are extracted in parallel in a single forward snapshot, completely bypassing the need for backpropagation. The physical silicon adapts to whatever non-linear transfer function actually exists.

15. Empirical Conclusions & Hardware Release

Definitive Empirical Conclusions:
1. **Naked Batch K=1 Benchmark:** For the first time, Inverted MAML has been demonstrated under single-shot vector evaluations without batch filtering, proving that closed-loop Vernier feedback suppresses raw silicon noise from 133,318 LSB² down to 19,028 LSB² (-85.7%).
2. **Resilience to Non-Stationary Disturbances:** The discrete 3D bar topographies confirm that FHP continuously reconfigures the spatial compensation landscape across physical crosspoints when subjected to severe ±5 LSB perturbations on Matrix 33.
3. **Hardware Ablation Verification:** Clamping weights to zero in Phase 4 instantly causes loss to jump by 7.7×, confirming that precision is sustained exclusively by the active 2-layer analog network.
Hardware Repository: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4